

Data Description Document

1. Data Sourcing and Licensing

The primary dataset used in this project is the **LIAR dataset**, a benchmark corpus for fake news and misinformation detection introduced by William Yang in the paper ‘*Liar, Liar Pants on Fire: A New Benchmark Dataset for Fake News Detection*’ (ACL 2017).

The dataset is publicly available for academic and research use and is hosted by the University of California Santa Barbara (UCSB) at:

https://www.cs.ucsb.edu/~william/data/liar_dataset.zip

It is released under an academic research license. There are no commercial use restrictions explicitly stated, but use must cite the original paper. No personally identifiable information (PII) is present in the dataset beyond speaker names and political affiliations of public figures, which are matters of public record.

2. Dataset Size and Language

The LIAR dataset contains **12,836 short statements** collected from PolitiFact, a public political fact-checking platform. The dataset is pre-split into three official partitions as shown in Table 1.

Table 1: Dataset split distribution (binary label mapping applied)

Split	Rows	Fake (0)	Real (1)
Train	10,269	5,193 (50.6%)	5,076 (49.4%)
Validation	1,284	645 (50.2%)	639 (49.8%)
Test	1,267	638 (50.4%)	629 (49.6%)
Total	12,820	6,476	6,344

Note: Binary label counts above reflect the mapping of the original 6-class labels as follows:

- **Fake (0):** Pants-fire, false, barely-true
- **Real (1):** Half-true, mostly-true, true

Language: All statements are in English. The corpus covers political speech primarily within the United States context.

Each record contains 14 fields including the claim text, fine-grained truthfulness label, speaker metadata (name, job title, party, state), credit history counts, and the context/venue of the statement.

3. Preprocessing Steps

All preprocessing is implemented in `src/liar_preprocessing.py` and is fully reproducible. The pipeline runs in six steps:

1. Data Loading

The three official `.tsv` files (`train.tsv`, `valid.tsv`, `test.tsv`) are loaded with fixed column names as defined by the LIAR schema. All fields are read as strings initially to prevent silent type coercion.

2. Missing Value Handling

Missing values are handled using a field-specific strategy:

- **Statement (critical field):** Rows where the statement is missing are dropped entirely. A model cannot learn without input text.
- **Credit history counts** (`barely_true_counts`, `false_counts`, etc.): Missing values are filled with 0. A missing count indicates the speaker has no recorded history on PolitiFact, which is semantically equivalent to zero.
- **Categorical fields** (speaker, party, subject, context, etc.): Missing values are filled with the string `"unknown"`. This preserves the row while signalling the absence of information to downstream models.

3. Text Cleaning

The `clean_text()` function applies the following transformations to the raw statement field:

- Strip leading and trailing whitespace
- Remove non-ASCII characters (e.g. smart quotes, accented characters from OCR noise)
- Collapse multiple consecutive whitespace characters into a single space
- Convert to lowercase

The cleaned output is stored in a new column `statement_clean`, preserving the original statement for reference.

4. Label Mapping

The original 6-class labels are mapped to two additional target columns:

- **label_binary:** Binary classification target (0 = fake, 1 = real)
- **label_ordinal:** Ordinal target from 0 (pants-fire) to 5 (true) for regression-style experiments

Any rows with unrecognised label values are logged and removed to prevent silent training errors.

5. Feature Engineering

The following features are derived to supplement the raw text for metadata-aware models:

Table 2: Engineered features and their rationale

Feature	Description	Rationale
<code>statement_len</code>	Word count of cleaned statement	Fake claims tend to be shorter and more assertive
<code>statement_char_len</code>	Character count	Complements word count for tokenisation-aware models
<code>exclamation_count</code>	Number of ‘!’ characters	Sensationalist language marker
<code>question_count</code>	Number of ‘?’ characters	Rhetorical questions common in misleading content
<code>capitalized_word_ratio</code>	Fraction of fully capitalised words	SHOUTING is a known misinformation signal
<code>speaker_lie_rate</code>	(false + pants-fire) / total statements for speaker	Speaker credibility history
<code>speaker_total_statements</code>	Total PolitiFact statements by speaker	Measures speaker prominence / data reliability
<code>is_political_context</code>	Binary flag for political venue keywords	Political context correlates with higher misinformation rate

6. Export

Processed splits are saved as UTF-8 encoded CSV files to `data/processed/` (`train_processed.csv`, `valid_processed.csv`, `test_processed.csv`). These files are excluded from version control via `.gitignore`. Raw `.tsv` files are never committed; they are fetched via `data/download_liar.py`.

4. Justification for Data Splits

The LIAR dataset provides official pre-defined train/validation/test splits. These pre-defined splits are used without modification for the following reasons:

- **Reproducibility:** Using the official splits allows direct comparison with published results (e.g., the Wang 2017 baseline, BERT-based replication studies).
- **No leakage risk:** The splits were defined by the dataset authors, ensuring no temporal or speaker-level leakage between partitions.
- **Community standard:** All major papers benchmarking on LIAR use these exact splits, making cross-paper comparison straightforward.

The approximately 80/10/10 split ratio (train/validation/test) is standard for NLP classification tasks and provides sufficient training data while reserving held-out sets for unbiased evaluation and hyperparameter selection respectively.

5. Known Limitations and Potential Biases

5.1. Source Bias (PolitiFact)

All statements are drawn from PolitiFact, a US-focused political fact-checking platform. This introduces several compounding biases:

- **Geographic bias:** The dataset is almost entirely US political speech. The system will generalise poorly to non-US political contexts or non-political domains (e.g. health misinformation, financial fraud).
- **Topic bias:** PolitiFact selects claims it deems newsworthy or politically significant. Mundane or non-political misinformation is entirely absent.
- **Selection bias:** Only claims that were submitted to and reviewed by PolitiFact fact-checkers are included. The fact-checking process itself introduces human editorial judgment.

5.2. Temporal Bias

The dataset was collected up to 2017. Political language, misinformation tactics, and social media discourse have evolved significantly since then. A model trained solely on LIAR may not detect modern misinformation formats (e.g. deepfake-associated claims, COVID-era health misinformation, AI-generated content).

5.3. Label Granularity and Subjectivity

The 6-class fine-grained labels (*pants-fire* $\rightarrow$ *true*) involve subjective human judgment. The boundary between *barely-true* and *half-true*, for instance, is not clearly defined and likely reflects annotator disagreement. The binary mapping used in this project collapses *half-true* into the “real” class, which may not be appropriate for all business use cases — a claim that is only half-true could be considered misleading in high-stakes settings.

5.4. Party and Speaker Imbalance

The dataset contains significantly more statements from Republican and Democrat politicians than from independents or third-party figures. Additionally, prominent speakers (e.g. Barack Obama, Donald Trump) are over-represented. This may cause the model to learn speaker identity as a proxy for truthfulness rather than generalising to content-based signals.

5.5. Class Balance at Fine-Grained Level

The binary label distribution is approximately balanced at 50.6% fake vs. 49.4% real across all splits, so class weighting is not required for the primary binary classifier used in this project. At the 6-class level, however, the *pants-fire* label is underrepresented relative to *half-true* and *mostly-true*. The utility module `src/class_weights.py` is provided to support experiments using the 6-class ordinal labels, where weighting or resampling would be appropriate.

5.6. Mitigation Strategies

The following mitigations are applied or planned within this project:

- Binary label aggregation reduces the impact of fine-grained label subjectivity.
- Speaker metadata features (`speaker_lie_rate`) are included to let models leverage credibility signals rather than memorising speaker identities.
- Class-weighted loss is available via `src/class_weights.py` for 6-class ordinal experiments. It is not applied to the primary binary classifier, as the binary label distribution is already balanced ($\sim 50/50$).
- Future work should augment the dataset with more recent sources (e.g. FEVER, MultiFC, or a curated social media corpus) to reduce temporal and domain bias.

References

Wang, W. Y. (2017). ‘Liar, Liar Pants on Fire’: A New Benchmark Dataset for Fake News Detection. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (ACL 2017)*, pp. 422–426.

LIAR dataset repository: https://www.cs.ucsb.edu/~william/papers/liar_dataset.zip